1. Clinical Question & Core Hypothesis

Clinical Question

Tier 1 synthesis highlights two robust but discordant findings — (A) noninvasive tumor-derived signals (urine methylation, EV mRNA/MDK) and immune infiltration are candidate biomarkers in NMIBC, and (B) metabolic marker DPYD is associated with poor prognosis. The key puzzle is: why do some TIL-rich NMIBC lesions still progress despite apparent immune infiltration?

Working Hypothesis

In non-muscle invasive bladder cancer (NMIBC), high infiltration of tumor-infiltrating lymphocytes (TIL) is not invariably associated with durable local control. We hypothesize that concurrent tumor-secreted extracellular vesicles (EVs) enriched for midkine (MDK) and/or high tumor DPYD (dihydropyrimidine dehydrogenase) activity metabolically and immunosuppressively reprogram the tumor microenvironment (TME). Specifically, high MDK-containing EVs and/or elevated DPYD activity will blunt effector T cell function and promote an immunometabolic milieu that supports tumor progression, even among TIL-rich NMIBC lesions. Consequently, NMIBC tumors with high TIL density but concomitant high MDK EV burden and/or high DPYD expression/activity will show shorter progression-free intervals (PFI) compared with TIL-rich tumors with low MDK EVs and low DPYD, and TIL density alone will have reduced prognostic value unless integrated with MDK and DPYD. The expanded, testable predictions are: (1) In NMIBC cohorts, TIL-high tumors with high MDK EV burden and/or high DPYD activity will have poorer PFI than TIL-high tumors with low MDK/DPYD, whereas TIL-low tumors with low MDK/DPYD will have the best PFI. (2) MDK-containing EVs derived from NMIBC cells suppress CD8+ T cell proliferation and cytotoxic function in vitro, an effect that is amplified in the presence of high DPYD activity; pharmacologic or genetic inhibition of MDK or DPYD will restore T cell activity. (3) In vivo, blocking MDK signaling or reducing DPYD activity will preserve or restore anti-tumor T cell function and delay progression in NMIBC models with TIL infiltration. (4) A composite biomarker integrating TIL density, EV-MDK burden, and DPYD activity will outperform TIL density alone for predicting PFI and guiding surveillance/therapy decisions.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MDK, DPYD
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MDK: MDK and metabolic enzyme DPYD) because these give a mechanistic axis linking tumor-secreted factors (EV-MDK) and tumor metabolism (DPYD) to modulation of local immune efficacy.

DPYD: DPYD) because these give a mechanistic axis linking tumor-secreted factors (EV-MDK) and tumor metabolism (DPYD) to modulation of local immune efficacy.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	Leukocyte_fraction	Supporting modifier from knowledge graph
Secondary	Immunoscore	Supporting modifier from knowledge graph
Secondary	EV_mRNA_biomarkers	Supporting modifier from knowledge graph
Secondary	Urine_DNA_methylation	Supporting modifier from knowledge graph
Secondary	EBV_DNA	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Leukocyte_fraction: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Immunoscore: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.